

CLASS 9 MATHEMATICS

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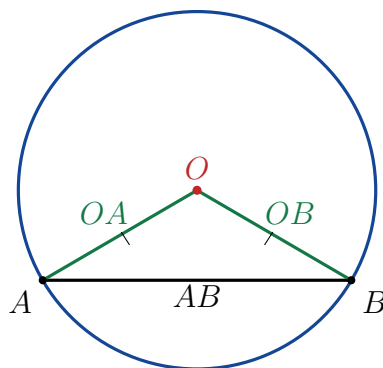
CHAPTER 5 – CIRCLES

EXERCISE 5.2

Question 1. Show that the triangle formed by a chord and the centre of the circle is isosceles.

Construction:

Let AB be a chord of a circle with centre O . Join OA and OB .



Proof:

Let AB be a chord of a circle with centre O .

Join OA and OB .

Since A and B lie on the circle,

$$OA = OB$$

because both OA and OB are radii of the same circle.

Therefore, in $\triangle OAB$,

$$OA = OB.$$

Hence, two sides of $\triangle OAB$ are equal.

Therefore,

$\triangle OAB$ is isosceles.

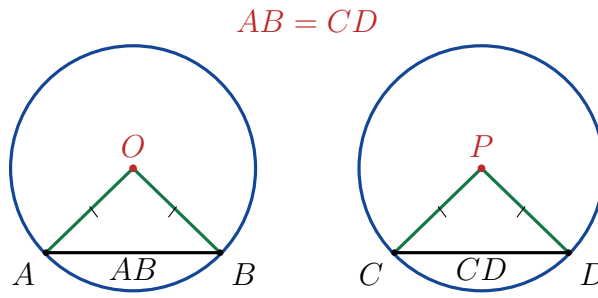
Question 2. Show that if two such isosceles triangles (occurring in the previous question) have equal base length, they are congruent to each other.

Construction:

Consider two circles with centres O and P . Let AB and CD be chords such that

$$AB = CD.$$

Join OA, OB, PC, PD .

**Proof:**

Let $\triangle OAB$ and $\triangle PCD$ be two isosceles triangles formed by joining the centre of a circle to the endpoints of a chord.

Since OA and OB are radii of the first circle,

$$OA = OB.$$

Similarly, PC and PD are radii of the second circle,

$$PC = PD.$$

Given that the base lengths are equal,

$$AB = CD.$$

Thus, the three corresponding sides are equal:

$$OA = PC, \quad OB = PD, \quad AB = CD.$$

Therefore, by the SSS congruence criterion,

$$\triangle OAB \cong \triangle PCD.$$

Hence,

$$\triangle OAB \cong \triangle PCD \quad \text{by SSS congruence.}$$

KEY IDEA

For a chord AB and centre O ,

$$OA = OB$$

because both are radii of the same circle.

Hence,

$$\triangle OAB \text{ is isosceles.}$$

If two such triangles have equal corresponding base lengths, then their three corresponding sides are equal, so

$$\text{SSS} \Rightarrow \text{Congruent triangles.}$$